PERSONAL INFORMATION

Al Shaima Al Rawahi

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al-shaima-al-rawahi

Astrophysics and Cosmology graduate with a strong research background and a passion for science education. Experienced in breaking down complex physical concepts—from exoplanet transits to stellar kinematics—for diverse audiences, including a public lecture for the Oman Astronomical Society. Eager to leverage hands-on experience with professional datasets (HST, TESS, Gaia) and computational methodologies to develop engaging, research-informed physics curricula and mentor the next generation of scientists.

EDUCATION

2023 – 2025

Master of Science in Astrophysics and Cosmology

105/110

University of Padova – Padova, Italy

- Relevant Pedagogy: Developed advanced communication skills by presenting complex thesis research on multiple stellar populations and black hole candidates to peers and faculty.
- Computational Focus: Gained proficiency in designing Python-based data pipelines, a skill directly transferable to creating structured lab exercises and computational physics projects for students.
- Thesis: *Multiple Stellar Populations in the Globular Cluster NGC 6440* – Analysis of HST WFC3/UVIS imaging to investigate cluster kinematics and internal dynamics.

2019 – 2022

Bachelor of Science in Physics with Astronomy

Second class division one

University of Liverpool – Liverpool, United Kingdom

- Thesis: *"Measuring the Shapes of Globular Clusters with Gaia"* – Independently processed Gaia DR3 data, wrote custom Python scripts, and produced a scientific report and presentation, demonstrating the ability to guide students through independent research projects.

TEACHING & OUTREACH EXPERIENCE

Invited Lecturer & Workshop Facilitator | *March 2026*
Oman Society of Astronomy and Space

- Designed and delivered an online workshop titled *"How We Discover What Cannot Be Seen: A Journey to Understand Black Holes and Star Clusters"* for society members and the public.

- Translated complex astrophysical concepts (light curves, stellar dynamics, black hole detection) into accessible, engaging content for a non-specialist audience.
- Received official certificate of appreciation from the Oman Astronomical Society for valuable contributions to public science education.

Graduate Research Mentor (Informal) | *2023 - 2025*
University of Padova

- Collaboratively worked with thesis advisors and fellow graduate students, effectively training peers in data-reduction techniques, Python-based analysis (PSF photometry, astrometric calibration), and the use of scientific software (Fortran tools, Astropy).
- Developed clear documentation and step-by-step workflows for data pipelines, mirroring the creation of robust lab manuals and instructional materials.

RESEARCH & LABORATORY EXPERIENCE

Note: The following experiences demonstrate my hands-on proficiency with the scientific method, data analysis, and technical problem-solving—skills I am eager to teach in a structured lab environment.

Hubble Space Telescope Data Analysis | *Thesis Project, 2025*
University of Padova

- Conducted an independent research project reducing and aligning HST WFC3/UVIS images to study stellar populations and kinematics in NGC 6440.
- Developed a comprehensive Python-based data pipeline for distortion correction, PSF photometry, and astrometric calibration, demonstrating the ability to manage complex, multi-step scientific workflows.
- Produced high-precision stellar catalogues and performed proper-motion analysis to search for black-hole candidates, an end-to-end research process that can be adapted to design advanced undergraduate lab exercises in data analysis and the scientific method.

Computational Astrophysics & Exoplanet Transit Analysis | 2024
University of Padova

- Applied supervised machine-learning (SVM, Neural Networks) and Bayesian MCMC fitting to detect exoplanetary transits and model planetary parameters.
- Gained expertise in designing, validating, and interpreting computational models—skills directly applicable to developing project-based curricula for courses in computational physics and data science.
- Maintained clear, reproducible code and documentation for all data-preprocessing and analysis steps (Python, scikit-learn, emcee, batman), mirroring best practices for creating structured lab manuals and instructional guides.

High-Energy Astrophysics Laboratory | 2023
University of Padova

- Performed UV and X-ray data reduction of the interacting galaxy system M51 using professional-grade pipelines (Swift UVOT/XRT, HEASoft, XSPEC).
- Modelled X-ray spectra to derive physical properties like photon index and luminosity, an experience that provides a foundation for teaching multi-wavelength astronomy and spectroscopic techniques.
- Collaborated effectively with lab partners to co-add images, produce light curves, and interpret complex datasets, simulating a team-based research environment suitable for upper-level physics labs.

Undergraduate Research — Globular Cluster Morphology with Gaia | 2022
University of Liverpool

- Independently processed Gaia DR3 data, writing custom Python scripts for cross-matching, statistical analysis, and data

visualization to quantify cluster ellipticities.

- Culminated the project in a scientific report and presentation, demonstrating the capacity to guide students through the full lifecycle of a research project—from hypothesis and data gathering to analysis and final communication.

Technical skills

Programming & Data Analysis	Astronomical Techniques	Software & Pipelines	Scientific Computation	Operating Systems
Python (NumPy, Matplotlib, Astropy, scikit-learn, emcee), Fortran (basic)	CCD Calibration, Aperture/PSF Photometry, Image Alignment, Time-Series Analysis	Astropy, wotan, batman, Git/GitHub, HEASoft, XSPEC	Bayesian inference (MCMC), Error Propagation, Model Fitting & Optimization	macOS / Linux (Bash scripting, conda environments)

KEY COMPETENCIES FOR TEACHING

- Communication: Proven ability to explain complex physical concepts to diverse audiences (public lecture, academic presentations).
- Mentorship: Experienced in guiding peers through technical workflows and data analysis.
- Curriculum Development: Skilled in structuring complex, multi-step research projects that can be adapted for undergraduate laboratory courses.
- Technical Writing: Proficient in producing clear scientific reports, documentation for data pipelines, and research summaries.

LANGUAGES

Arabic Mother tongue

English Proficient

الجمعية العُمانية للفلك والفضاء
Oman Society of Astronomy and Space

عُمان
OMAN

شهادة شكر وتقدير

تتقدم الجمعية العُمانية للفلك والفضاء بخالص الشكر والتقدير للفاضلة/

الشيماء بنت عمرو الرواحية

وذلك نظير جهودها القيّمة في تقديم ورشة "كيف نكتشف ما لا يُرى: رحلة لفهم الثقوب السوداء والعناقيد النجمية"، والتي عُقدت عن بُعد يوم الأربعاء بتاريخ ١٥ رمضان ١٤٤٧ هـ الموافق له ٤ مارس ٢٠٢٦ م. متمنين لها دوام التوفيق والمشاركة الفعالة في مسيـرة الجمعية العُمانيـة للفـلك والفضـاء.

د. إسحاق بن يحي الشعيلي
رئيس مجلس الإدارة

الجمعية العمانية للفلك والفضاء
OMAN SOCIETY OF ASTRONOMY & SPACE

REPUBBLICA ITALIANA IN NOME DELLA LEGGE

NOI PROFESSORESSA DANIELA MAPELLI RETTRICE DELLA

UNIVERSITÀ DEGLI STUDI DI PADOVA

visti gli attestati degli studi compiuti
visto il risultato dell'esame di laurea superato il giorno 10 dicembre 2025

CONFERIAMO A

AL SHAIMA AMRO MOHAMMED AL RAWAHI

nata a Suhar (Oman) il 23 settembre 2000

LA LAUREA MAGISTRALE IN

Astrophysics and Cosmology

classe LM-17 delle lauree magistrali in Fisica (ex DM 270/04)

Padova, addì 10 dicembre 2025

Reg. n. 407730

La Rettrice

Il Direttore Generale

MCCXXII UNIVERSA UNIVERSIS PATAVINA LIBERTAS MCCXXII UNIVERSA UNIVERSIS PATAVINA LIBERTAS MCCXXII

imposta di bollo assolta in modo virtuale (aut. Int. di Finanza - Pd del 20/02/1991 n°4443/91/2t)

UNIVERSITY OF LIVERPOOL

AWARD CERTIFICATE

DIGITAL VERSION

Faculty of Science and Engineering

Degree of Bachelor of Science with Honours

We hereby certify that on the authority of the senate

Al Shaima Amro Mohammed Al Rawahi

was awarded the degree of *Bachelor of Science with Honours in Physics with Astronomy*

(Class Two Division One)

on the fourth day of July 2022 and that the degree was conferred at a ceremony on the twenty-second day of July 2022

Vice Chancellor

UNIVERSITY OF LIVERPOOL